1.5 V Fully Differential Class-AB Operational Amplifier

Architectural & Subcircuit Documentation

1. Architectural Overview & Introduction

Block #	Subcircuit / Module Name	Primary Architectural Role
Block 1	ota_cmfb_top_tb	Top-Level Multi-Mode SPICE Testbench Schematic
Block 2	ota_cmfb_top	Top-Level Structural Integration, CMFB Sensing & ESD/Antenna Guarding
Block 3	ota_cmfb_core	5-Transistor PMOS Differential Input Core Stage & Output Stage Wrapper
Block 4	ota_cmfb_core_push_pull_output_stage	Monticelli Class-AB Output Driver with $g_{mp} >> g_{mn}$ Decoupling
Block 5	ota_cmfb_fbota	Continuous-Time Common-Mode Feedback (CT-CMFB) Error Amplifier
Block 6	ac_diff_probe	Differential AC Probe for Open-Loop Gain and Phase Measurement
Block 7	ac_probe	Single-Ended AC Loop-Breaking Probe

2. Block 1: Top-Level Testbench (ota_cmfb_top_tb)

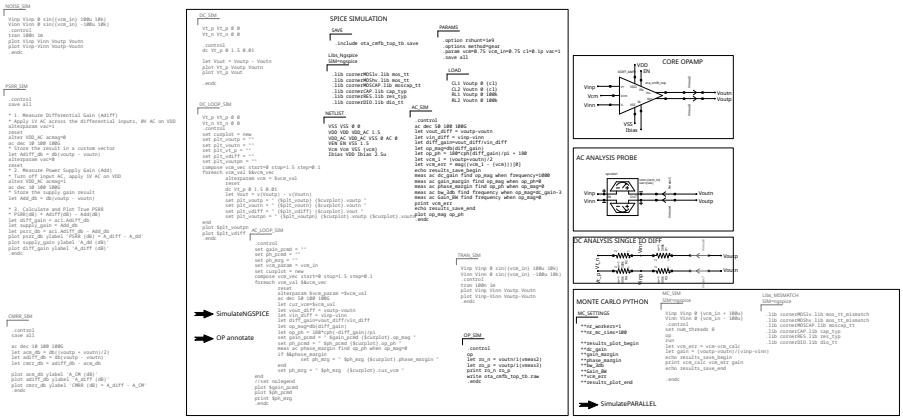


Figure 1: *Top-Level Testbench Structured Schematic Diagram (ota_cmfb_top_tb)*

Functional Description & Role: Executes all required DC, AC, transient, noise, PSRR, CMRR, and statistical Monte Carlo mismatch simulations for the operational amplifier macro.

Key Technical Specifications:

- **Modular Simulation Control:** Toggle individual simulation blocks (OP_SIM, DC_SIM, DC_LOOP_SIM, AC_SIM, AC_LOOP_SIM, TRAN_SIM, NOISE_SIM, PSRR_SIM, CMRR_SIM, MC_SIM) on or off and click SimulateNGSPICE or SimulatePARALLEL to launch.
- **Unified Verification Environment:** Consolidates nominal corner verification (mos_tt, moscap_tt, res_typ, dio_tt) and 100-run Monte Carlo mismatch analysis (mos_tt_mismatch) within a single testbench interface with default load conditions C_L = 0.1 pF and R_L = 100 kΩ.

3. Block 2: Top-Level Integration (ota_cmfb_top)

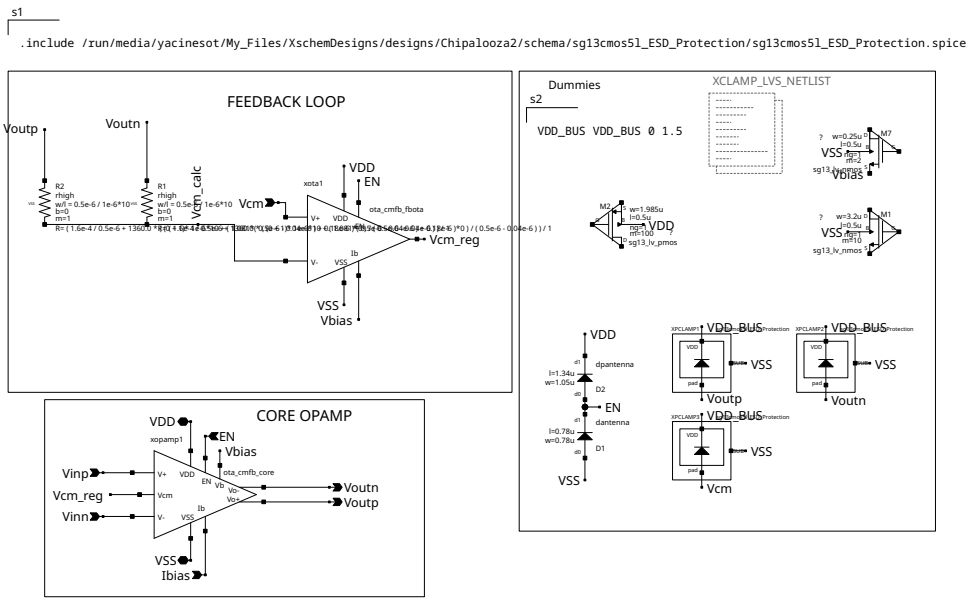


Figure 2: *Top-Level Operational Amplifier Structural Schematic Diagram (ota_cmfb_top)*

Functional Description & Role: Integrates the core amplifier (xopamp1), continuous-time CMFB error amplifier (xota1), resistive common-mode sensing divider (R1, R2), layout dummy devices, and global ESD/antenna protection into a single macro pinout.

Key Technical Specifications & Component Parameters:

- **Master Supply Rails:** VDD = 1.5 V, VSS = 0.0 V.
- **Control & Reference Inputs:** Vinp, Vinn, Vcm (output common-mode target range: 0.3 V to 1.1 V), EN (enable), Ibias (2.5 µA reference current).
- **Common-Mode Sense Network:** Matched resistive divider (R1 = R2, W/L = 0.5 µm / 10.0 µm, m = 1) using the high-resistivity polysilicon resistor model (rhigh) to sense the output common-mode voltage $V_{cm,calc} = (V_{outp} + V_{outn}) / 2$, sized for optimal matching and low thermal noise.
- **Differential Outputs:** Voutp, Voutn (1.2 V_{pp} differential swing).
- **Antenna Protection Circuitry:** Reverse-biased antenna clamp diodes D1 (dantenna, W = 0.78 µm, L = 0.78 µm) and D2 (dpantenna, W = 1.05 µm, L = 1.34 µm) on the EN net.
- **Additional ESD Clamping:** Because this circuit is integrated into a multi-project wafer (MPW) where the PDK's analog I/O pads clamp input voltages to V_{high} = 3.3 V and VSS, dedicated secondary ESD clamps (XPCLAMP1–XPCLAMP3, sg13cmos5l_ESD_Protection) are added on Voutp, Voutn, and Vcm to clamp voltages locally to the 1.5 V core supply rail (VDD_BUS).

4. Block 3: Differential Input Core Stage (ota_cmfb_core)

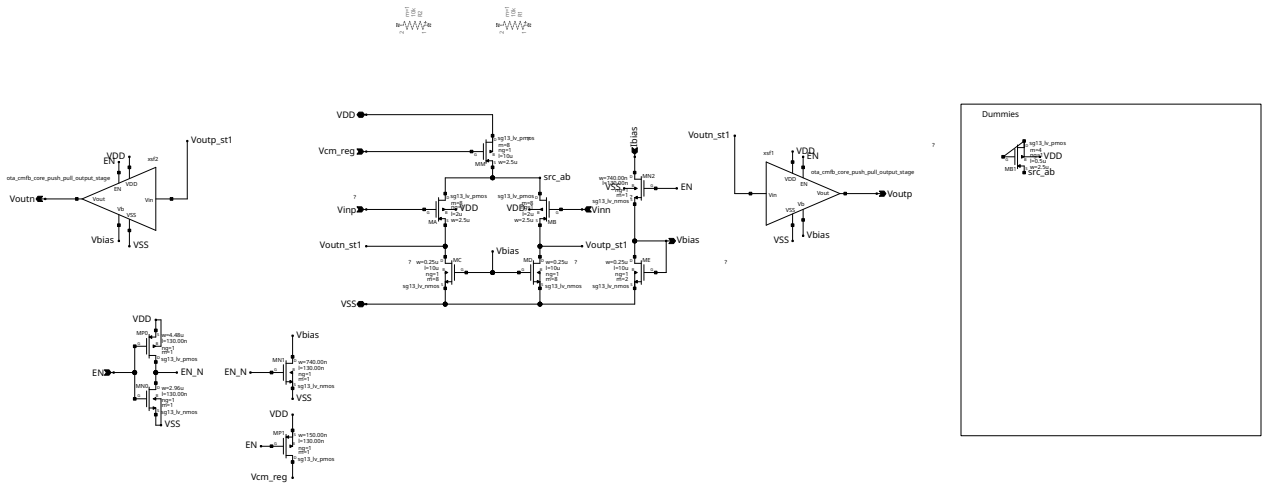


Figure 3: *5T PMOS Differential Pair Core & Output Stage Wrapper Schematic (ota_cmfb_core)*

Functional Description & Role: Provides primary input transconductance (g_{m1}) and first-stage voltage gain (A_{v1}) with ultra-low flicker (1/f) noise, driving the dual Class-AB push-pull output stage instances (xsf1, xsf2).

Key Technical Specifications & Transistor Parameters:

- **Input Drivers (MA / MB):** PMOS differential pair (sg13_lv_pmos) sized W/L = 2.5 µm / 2.0 µm (m = 8) to suppress 1/f flicker noise and maximize intrinsic device gain ($g_m \cdot r_{out}$).
- **Active Loads (MC / MD):** Long-channel NMOS active loads (sg13_lv_nmos, W/L = 0.25 µm / 10.0 µm, m = 8) biased via diode-connected reference transistor ME (W/L = 0.25 µm / 10.0 µm, m = 2) to provide high first-stage output impedance.
- **Tail Bias Generator (MM):** PMOS current source (sg13_lv_pmos, W/L = 2.5 µm / 10.0 µm, m = 8) controlled directly by the CMFB error signal (Vcm_reg).
- **First-Stage Output Nodes:** Voutn_st1 and Voutp_st1 driving the push-pull output stage instances (xsf1, xsf2).

5. Block 4: Monticelli Class-AB Output Stage (ota_cmfb_core_push_pull_output_stage)

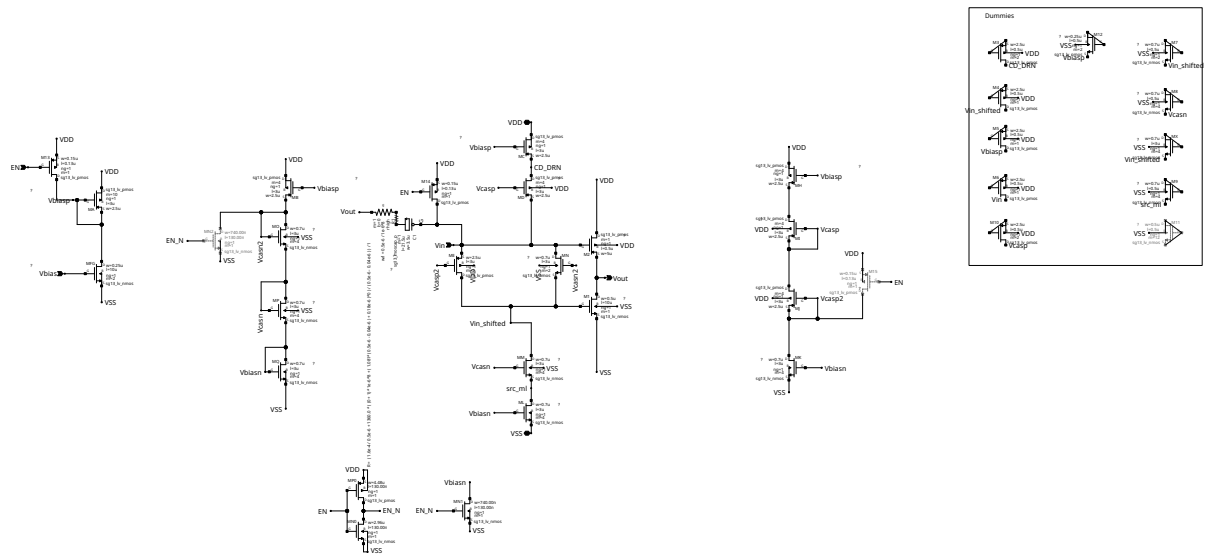


Figure 4: *Class-AB Output Stage & Miller Compensation Schematic (ota_cmfb_core_push_pull_output_stage)*

Functional Description & Role: A fully self-contained, cascoded Monticelli Class-AB push-pull output buffer providing dynamic current drive, wide rail-to-rail output swing ($0.1\,\mathrm{V}$ to $1.3\,\mathrm{V}$), and second-stage voltage gain. The block integrates its own internal cascode and translinear mesh bias generators derived directly from the $2.5\ \mu\mathrm{A}$ master reference (Vbias), along with dedicated active-high enable (EN) tie-off switches on every floating bias and signal node.

Key Technical Specifications & Transistor Parameters:

- **Primary Reference Mirror (MFG / MA):** MFG (sg13_lv_nmos, $W/L = 0.25\ \mu\mathrm{m}/10.0\ \mu\mathrm{m}$, $m = 2$) is unit-matched to the first-stage bias reference (ME) to accurately copy the $2.5\ \mu\mathrm{A}$ reference current, driving diode-connected PMOS load MA (sg13_lv_pmos, $W/L = 2.5\ \mu\mathrm{m}/3.0\ \mu\mathrm{m}$, $m = 10$) to establish the master PMOS bias rail (Vbiasp).
- **Cascoded Current-Source Loads (MC, MD & ML, MM):** In textbook Monticelli topologies, the floating Class-AB control mesh is biased by ideal current sources; here, high-impedance cascoded current sources are implemented via MC (main PMOS current source) cascoded by MD (both $W/L = 2.5\ \mu\mathrm{m}/3.0\ \mu\mathrm{m}$, $m = 4$) on the upper branch, and ML (main NMOS current source) cascoded by MM (both $W/L = 0.7\ \mu\mathrm{m}/3.0\ \mu\mathrm{m}$, $m = 4$) on the lower branch.
- **Floating Class-AB Mesh / Level Shifters (ME & MN):** Head-to-tail translinear mesh transistors ME (sg13_lv_pmos, $W/L = 2.5\ \mu\mathrm{m}/3.0\ \mu\mathrm{m}$, $m = 2$, biased by Vcasp2) and MN (sg13_lv_nmos, $W/L = 0.7\ \mu\mathrm{m}/3.0\ \mu\mathrm{m}$, $m = 2$, biased by Vcasn2) act as floating voltage level shifters between the Vin (PMOS gate) and Vin_shifted (NMOS gate) nodes, regulating quiescent Class-AB current while providing dynamic push-pull drive.
- **Integrated Cascode & Mesh Bias Generators:** Two multi-device replica stacks generate all required cascode and mesh bias potentials locally:PMOS Bias Branch (MH, MI, MJ —

$W/L = 2.5\ \mu\text{m}/3.0\ \mu\text{m}$, $m = 4$): Establishes Vcasp and Vcasp2, terminated at the bottom by NMOS diode load MK ($W/L = 0.7\ \mu\text{m}/3.0\ \mu\text{m}$, $m = 4$) to generate Vbiasn.

- **NMOS Bias Branch (MB, MO, MP, MQ):** Mirrors the PMOS current via MB ($W/L = 2.5\ \mu\text{m}/3.0\ \mu\text{m}$, $m = 4$) into the NMOS stack (MO, MP, MQ — $W/L = 0.7\ \mu\text{m}/3.0\ \mu\text{m}$, $m = 4$) to generate Vcasn2 and Vcasn. Pairing MK and MQ with MH and MC enforces strict current matching across all branches.
- **Asymmetric Output Drivers (M2 & M1) & Pole-Zero Positioning:** PMOS driver M2 ($W/L = 5.0\ \mu\text{m}/0.5\ \mu\text{m}$, $m = 1$) and long-channel NMOS driver M1 ($W/L = 0.5\ \mu\text{m}/10.0\ \mu\text{m}$, $m = 1$) establish $g_{mp} \gg g_{mn}$ asymmetry, pushing the NMOS right-half-plane (RHP) zero well above the target unity-gain bandwidth ($\text{GBW} = 40 - 50\,\text{MHz}$).
- **Miller Compensation Network (C1 & R1):** PMOS accumulation capacitor C_C (C1, sg13_moscap_p, $W = 3.5\ \mu\text{m}, L = 21.5\ \mu\text{m}, m = 1$) in series with nulling resistor $R_C = 23.97\,\text{k}\Omega$ (R1, rhigh, $W/L = 0.5\ \mu\text{m}/8.0\ \mu\text{m}, m = 1$) guarantees a $60\ -65\ $ phase margin.
- **Full Floating-Node Enable / Power-Down Protection:** An active-high enable (EN) inverter (MP0 / MN0) generates EN_N to drive dedicated pull-up (M13, M14, M15 to VDD) and pull-down (MN1, MN2 to VSS) switches on every internal bias and floating node (Vbiasp, Vbiasn, Vcasp2, Vcasn2, Vin), eliminating floating-gate leakage and ensuring deterministic zero-current shutdown when EN = 0.

6. Block 5: Continuous-Time CMFB Amplifier (ota_cmfb_fbota)

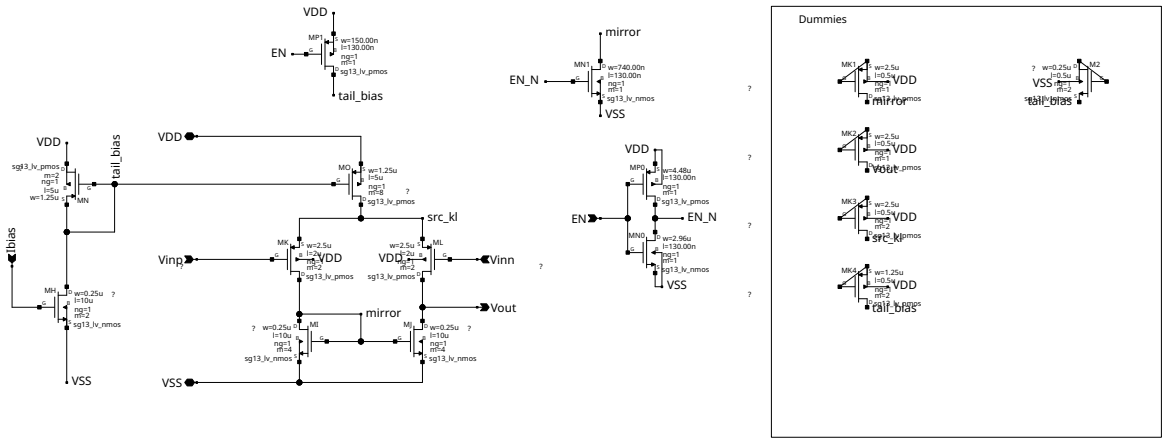


Figure 5: *Continuous-Time Common-Mode Feedback Error OTA Schematic (ota_cmfb_fbota)*

Functional Description & Role: Senses the output common-mode voltage error and regulates the main core tail bias (Vcm_reg) to lock $V_{cm,out}$ at the user's reference input Vcm.

Key Technical Specifications & Transistor Parameters:

- **Error Input Pair (MK / ML):** PMOS differential input pair (sg13_lv_pmos, W/L = 2.5 µm / 2.0 µm, m = 2) comparing the sensed common-mode voltage (Vcm_calc) against the Vcm reference.
- **Active Mirror Load (MI / MJ):** Long-channel NMOS current mirror load (sg13_lv_nmos, W/L = 0.25 µm / 10.0 µm, m = 4) providing the single-ended regulation output Vout (Vcm_reg).
- **Tail Current Mirror (MO / MN / MH):** PMOS tail current source MO (W/L = 1.25 µm / 5.0 µm, m = 8) mirrored from diode-connected MN (m = 2) and reference sink MH (W/L = 0.25 µm / 10.0 µm, m = 2).
- **Low-Frequency Common-Mode Rejection:** Provides >97.5 dB CMRR at low frequencies under statistical Monte Carlo mismatch.

7. Block 6: Differential AC Probe (ac_diff_probe)

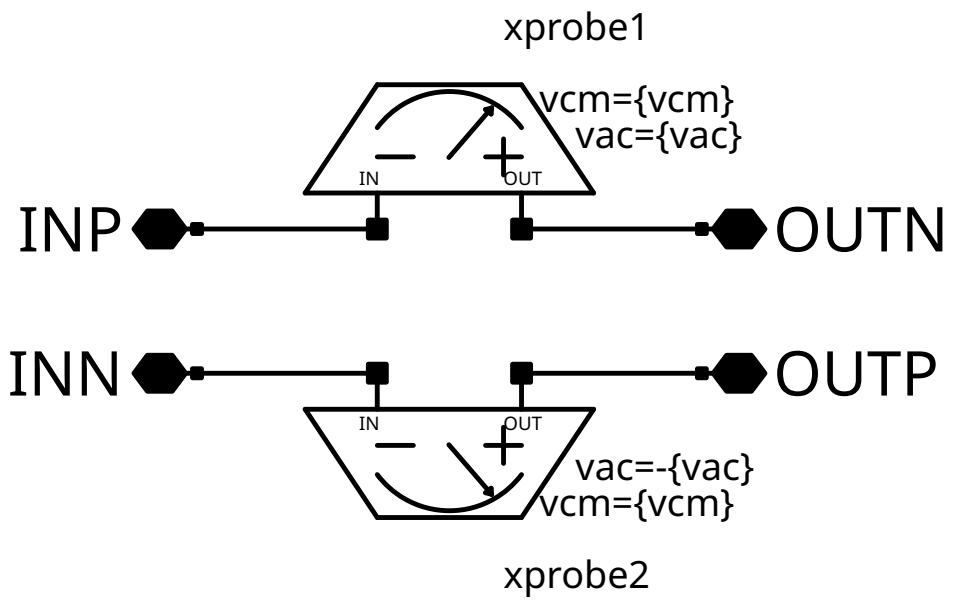


Figure 6: *Differential AC Probe Schematic (ac_diff_probe)*

Functional Description & Role: Closes the operational amplifier feedback loop at DC while blocking AC feedback and injecting a differential AC stimulus to measure open-loop gain and phase in a fully differential topology.

Key Technical Specifications:

- **Dual Single-Ended Probe Topology:** Instantiates two matched single-ended AC probes (xprobe1 driven with vcm={vcm}, vac={vac} between INP and OUTN; xprobe2 driven with anti-phase vcm={vcm}, vac=-{vac} between INN and OUTP).

8. Block 7: Single-Ended AC Probe (ac_probe)

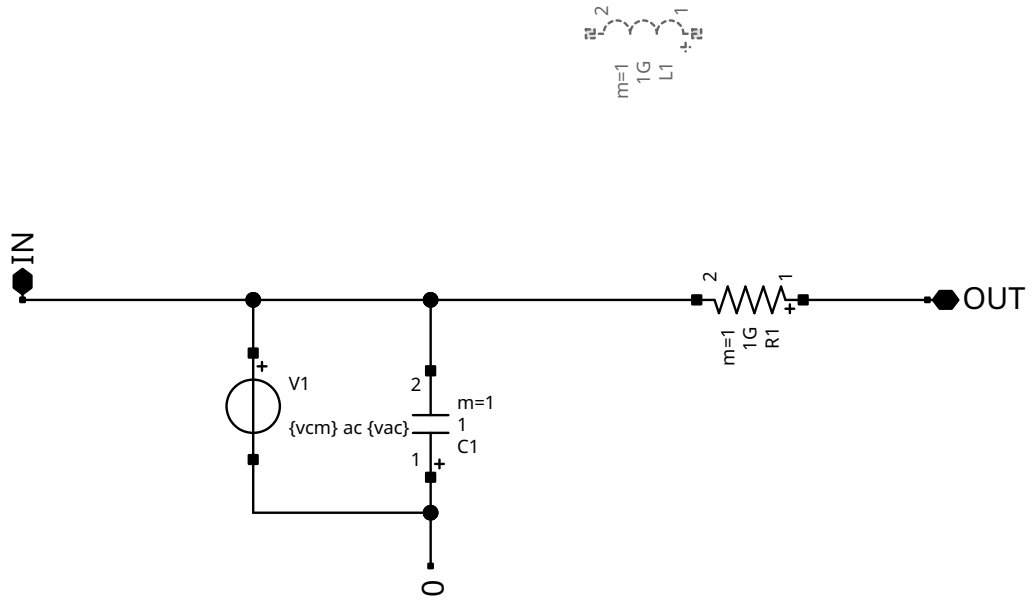


Figure 7: *Single-Ended AC Probe Schematic (ac_probe)*

Functional Description & Role: Used as the core building block inside the differential AC probe (ac_diff_probe), and can also be used standalone in single-ended amplifier topologies to sense open-loop AC gain while maintaining DC operating-point feedback.

Key Technical Specifications & Component Parameters:

- **Large-RC Loop-Breaking Network:** Combines a 1 GΩ feedback resistor (R1 = 1G, m = 1) between OUT and IN with a 1 F AC-shunt capacitor (C1 = 1, m = 1) and parameterized AC/DC voltage source V1 ({vcm} ac {vac}) to pass DC feedback while isolating the AC test signal.